Nujoum Unus

Data Scientist

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PROFESSIONAL SUMMARY

Data scientist (MS Business Analytics, UC San Diego) who builds **predictive models, experiments and data pipelines** in SQL, Python and PySpark, from a LightGBM fraud model sized against false-positive costs and a credit risk model for a consumer lender to uplift models on randomized experiments.

WORK EXPERIENCE

BriteCap Financial

San Diego, CA | Apr 2025 - Jun 2025

Data Science Consultant

- Built and validated a Propensity-to-Pay model (Logistic Regression, Random Forest, **LightGBM**) with lagged features against label leakage, KS and PSI stability screening and a borrower-level train/test split
- Engineered behavioral features from **24M+ Plaid bank transactions** across 7,600+ loan applications and explained drivers with SHAP
- Segmented customers by financial trend and funnel outcome to target outreach, and presented findings and a reusable pipeline to leadership

TATA ELXSI

India | Nov 2023 - May 2024

Automation Engineer

- Built a GenAI prototype with LLMs and AWS SageMaker to automate UML documentation, **cutting effort ~60%** and winning a top-three company innovation award
- Ranked **#1 of 60+ trainees** as a "Rising Star" for rapid onboarding, problem-solving and cross-functional collaboration

PROJECTS

Credit Card Fraud Detection Model

Python, LightGBM

Built a LightGBM fraud model on 98K+ card transactions, catching **77% of out-of-time fraud at an 8% review rate**; sized cutoffs against false-positive costs (~$56.7M est. net annual savings, over 99% of the maximum).

Uplift & Propensity Targeting (Creative Gaming)

Python, Logistic Regression, Random Forest, Uplift Modeling

Built uplift models on **randomized ad-experiment data** to target users an ad actually persuades, selecting the best 30,000 of 120,000 users to maximize incremental profit.

Distributed Data Pipelines with PySpark & AWS EMR

PySpark, Spark SQL, HDFS, S3, EMR, SageMaker

Built PySpark ETL on HDFS and AWS EMR over **18.5M records** and deployed a tuned XGBoost model to a SageMaker endpoint.

EDUCATION

Master of Science in Business Analytics (GPA 3.6/4) University of California, San Diego — Aug 2024 - Dec 2025

Bachelor of Science in Computer Science and Engineering (GPA 3.58/4) A.P.J. Abdul Kalam Technological University — Jul 2019 - Jun 2023

SKILLS

Analytics: SQL, Python (pandas, NumPy, statsmodels, scikit-learn), A/B testing, experimental design, uplift modeling, forecasting, statistics

Tools & ML: PySpark, Snowflake, PostgreSQL, AWS (EMR, SageMaker), Tableau, LightGBM, XGBoost, LLMs, MCP agents, Git